

L07 VISUAL BALANCE | O (MAX : 1 PT)

Intent : Create lighting environments that enhance visual comfort.

Summary : This WELL feature requires projects to develop and implement strategies to create a visually comfortable lighting environment.

Issue : A lighting environment may utilize different kinds of light sources, including daylighting and different types of electric lighting technology. Lighting is planned across spaces to achieve the required ambience and energy efficiency goals. However, fluctuating light levels impact the visual comfort of occupants and could lead to eye fatigue.¹ Sudden increases and decreases in brightness can cause a high level of visual discomfort.² Constant changes in lighting have been linked to distraction from the task at hand, which can decrease productivity and impact well-being.³

Solutions : Development of a lighting layout and operations schedule to complement the lighting design in a space is key to increasing the comfort of users. Evidence suggests that thoughtful planning of lighting in a space that takes into account color temperature, daylight and electric light supports a visually comfortable lighting environment.^{4,5} Consideration of the ages of users, tasks performed and existing physical features in the space are also integral to creating a productive space.

PART 1 BALANCE VISUAL LIGHTING (MAX : 1 PT)

For All Spaces:

Option 1: Parameters for visual balance

Ambient lighting in all regularly occupied spaces meets at least three of the following requirements:

- a. Horizontal and vertical luminance contrast ratios for an ambient light system is no more than 10:1 between adjacent independently controlled zones.
- b. Illuminance uniformity ratio of at least 0.4 or 1:2.5 (minimum light level: average light level) is achieved on any horizontal task plane within a space.
- c. One of the following:
 1. A lighting automation system is in use and automatic changes in lighting characteristics, such as light levels, changes in color and distribution take place over a period of at least 10 minutes.
 2. A lighting automation system is not in use.
- d. The Correlated Color Temperature (CCT) in each room for similar fixtures is consistent (± 200 K) at any point of time.

OR

Option 2: Design for visual balance

Lighting is designed by a lighting professional and takes into account the following considerations:

- a. Luminance ratios on vertical and horizontal adjacent zones.
- b. Illuminance uniformity on horizontal task planes.
- c. Changes in lighting characteristics, such as light levels, changes in color and distribution.
- d. Correlated Color temperature (CCT) of lights used.

REFERENCES

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